

## Cauchy's Theorem

The difference between actual function value and polynomial function is -

$$\begin{aligned} \text{Error} &= |f(x) - P_n(x)| \\ &= \frac{f^{(n+1)}(\xi) (x-x_0)(x-x_1)\dots(x-x_n)}{(n+1)!} \end{aligned}$$

Say

Function =

↳ 1<sup>st</sup> term, 2<sup>nd</sup> term, stop  
3<sup>rd</sup> term, ..., n<sup>th</sup> term,  
n+1<sup>th</sup> term, n+2 term.  
error part.

Now, we need to find Maximum Error  
or, upper bound Error

Ref. 1-3 Rec.

A function is given,

and  $\xi \in [-1, 1]$

and,  $x \in \{\frac{\pi}{2}, 0, \pi\}$

$$f(x) = \cos(x)$$

so,

| $x$                       | $f(x) \rightarrow \cos(x)$ |
|---------------------------|----------------------------|
| $x_0 \dots \frac{\pi}{2}$ | 0                          |
| $x_1 \dots 0$             | 1                          |
| $x_2 \dots \pi$           | -1                         |

$P_2(x)$

$$\therefore \text{Error} = |f(x) - P_2(x)|$$

$$= \frac{f^3(\xi)(x-x_0)(x-x_1)(x-x_2)}{3!}$$

need to find the MAX value of these terms.

$$= \underbrace{\left| \frac{\sin(\xi)}{3!} \right|}_{\xi \text{ part}} \cdot \underbrace{\left| (x - \frac{\pi}{2})(x-0)(x-\pi) \right|}_{w(x) \text{ part}}$$

Here,

$$\begin{aligned} f(x) &= \cos(x) \\ f'(x) &= -\sin(x) \\ f''(x) &= -\cos(x) \\ f'''(x) &= \sin(x) \end{aligned}$$

# Max value of  $\xi$  part:

given,  $\xi \in [-1, 1]$

for  $-1$ , we get  $\sin(-1) = -0.84147$

for  $1$ , we get  $\sin(1) = 0.84147$  max!

So,  $\left| \frac{\sin(\xi)}{3!} \right| = \left| \frac{\sin(1)}{3!} \right| = \frac{0.84147}{3!} = 0.14$

Max  
or  $\xi$  part

# Let's find for Max  $w(x)$  part:

$$w(x) = \left| (x - \frac{\pi}{2})(x-0)(x-\pi) \right|$$

— Doing  $w'(x) = 0$  to get max value,

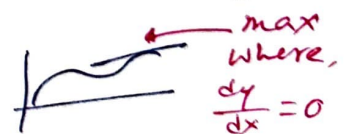
So,  $w'(x) = 0$

$$\Rightarrow 3x^2 - 3\pi x + \frac{\pi^2}{2} = 0$$

$$\therefore x = \frac{\pi}{2} \left( 1 \pm \frac{1}{\sqrt{3}} \right)$$

Note!!

To get maximum value/position of any function. Do  $\Rightarrow \frac{dy}{dx} = 0$



plug the  $x$  and  $\xi$  values in  $w(x)$ :  $\rightarrow [-1, 1]$

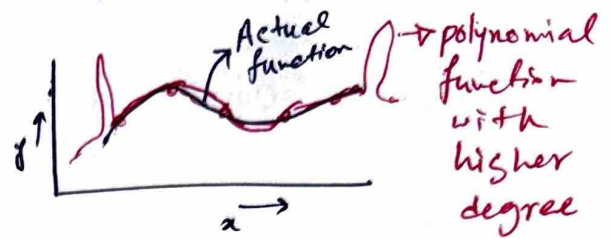
| $x$                                       | $w(x) =  (x - \frac{\pi}{2})(x - 0)(x - \pi) _{\max}$ |
|-------------------------------------------|-------------------------------------------------------|
| $\frac{\pi}{2}(1 + \frac{1}{\sqrt{3}})$   | $ -1.4919  = 1.4919$                                  |
| $\frac{\pi}{2}(1 - \frac{1}{\sqrt{3}})$   | $ 1.4919  = 1.4919$                                   |
| $\xi$ $\begin{cases} -1 \\ 1 \end{cases}$ | $ -10.648  = 10.648 \rightarrow \text{MAX!}$          |
|                                           | $ 1.22  = 1.22$                                       |

$$\begin{aligned}
 \therefore \text{Upper bound Error / Max Error} &= \xi_{\text{part}} \times w(x)_{\text{part}} \\
 &= 0.14 \times 10.648 \\
 &= \boxed{1.49072}
 \end{aligned}$$

Ans.

## Runge Phenomenon & Chebyshev's Node

After plotting polynomial function, we get error at corner edges, this concept is called **Runge Phenomenon**.



### Reasons of occurring Runge Phenomenon:

- Number of degree and internal spaces.
- Function we are getting.

Possible sol<sup>n</sup>:

- ignore equal spacing between nodes.
- increase number of nodes at corner edges.

To do this we need **Chebyshev's Nodes**.

(Not equal distanced)

Proof

$$\phi_j = \frac{(2j+1)\pi}{2(n+1)}$$

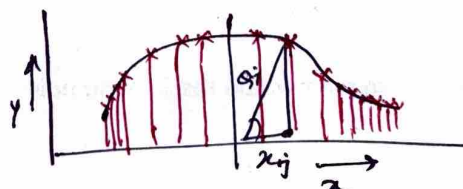
$$\frac{x_j}{\cos \phi_j} = \frac{x_1}{1}$$

$$\therefore x_j = \cos \phi_j$$

$$x_j = \cos \frac{(2j+1)\pi}{2(n+1)}$$

**Chebyshev's nodes formula.**

Note: Runge Phenomenon doesn't work on every function. It varies on function, nodes etc.



the proof of formula given is Rec!

Ex.

Runge Function,  $f(x) = \frac{1}{16x^2}$  ; the interval  $[-1, 1]$   
degree,  $n = 2$

Applying Formula:

$$x_0 = \cos\left(\frac{(2 \times 0 + 1)\pi}{2(2+1)}\right) = \cos\frac{\pi}{6}$$

$$x_1 = \cos\frac{(2 \times 1 + 1)\pi}{2(2+1)} = \cos\frac{3\pi}{6}$$

$$x_2 = \cos\frac{(2 \times 2 + 1)\pi}{2(2+1)} = \cos\frac{5\pi}{6}$$

now you have  
 $x_0, x_1, x_2$  points/  
nodes.  
you can use them now.

you got your  
table. Now you can  
do any **Interpolation**.

| $x$                            | $f(x)$ |
|--------------------------------|--------|
| $x_0 \dots \cos\frac{\pi}{6}$  |        |
| $x_1 \dots \cos\frac{3\pi}{6}$ |        |
| $x_2 \dots \cos\frac{5\pi}{6}$ |        |